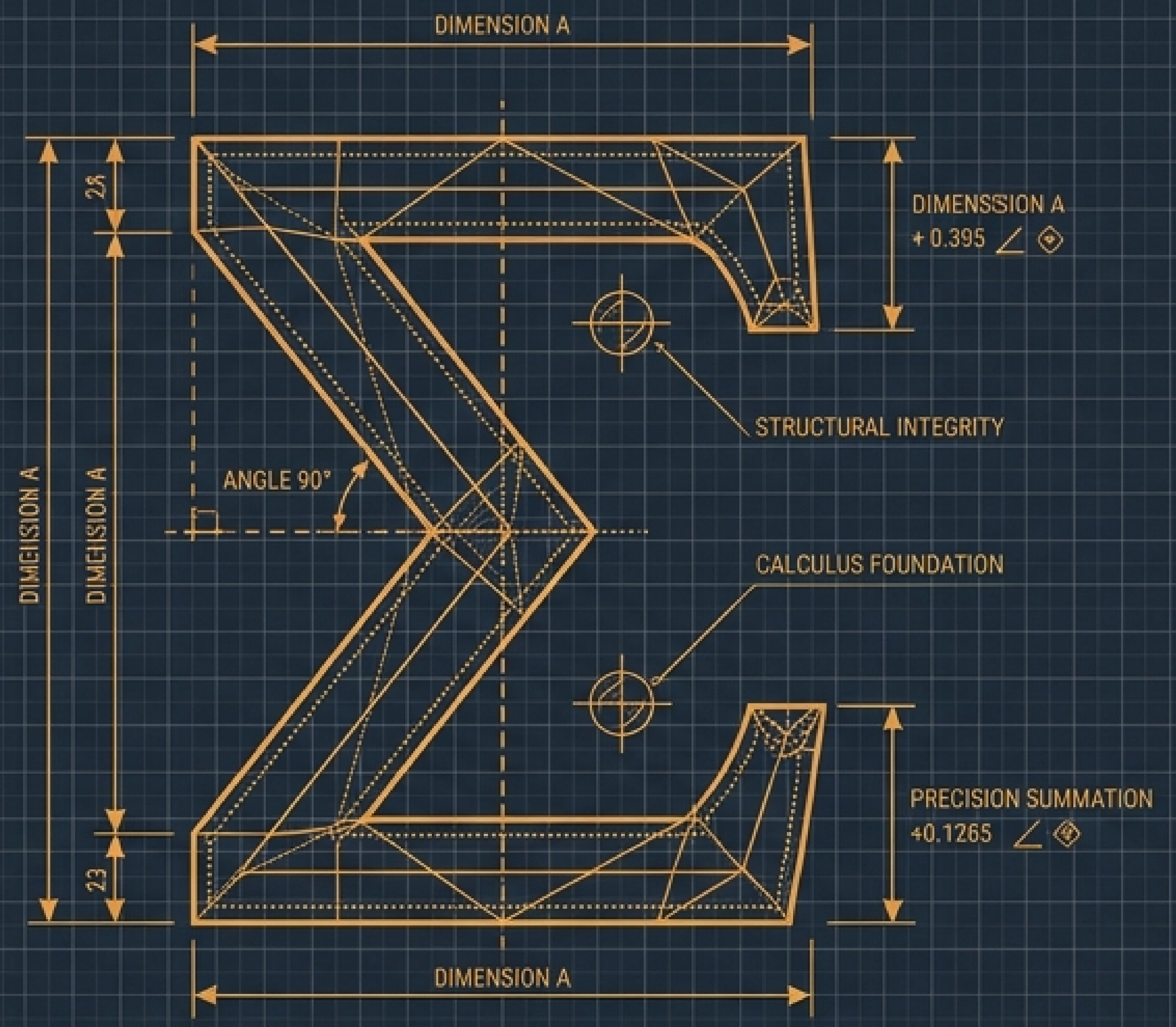
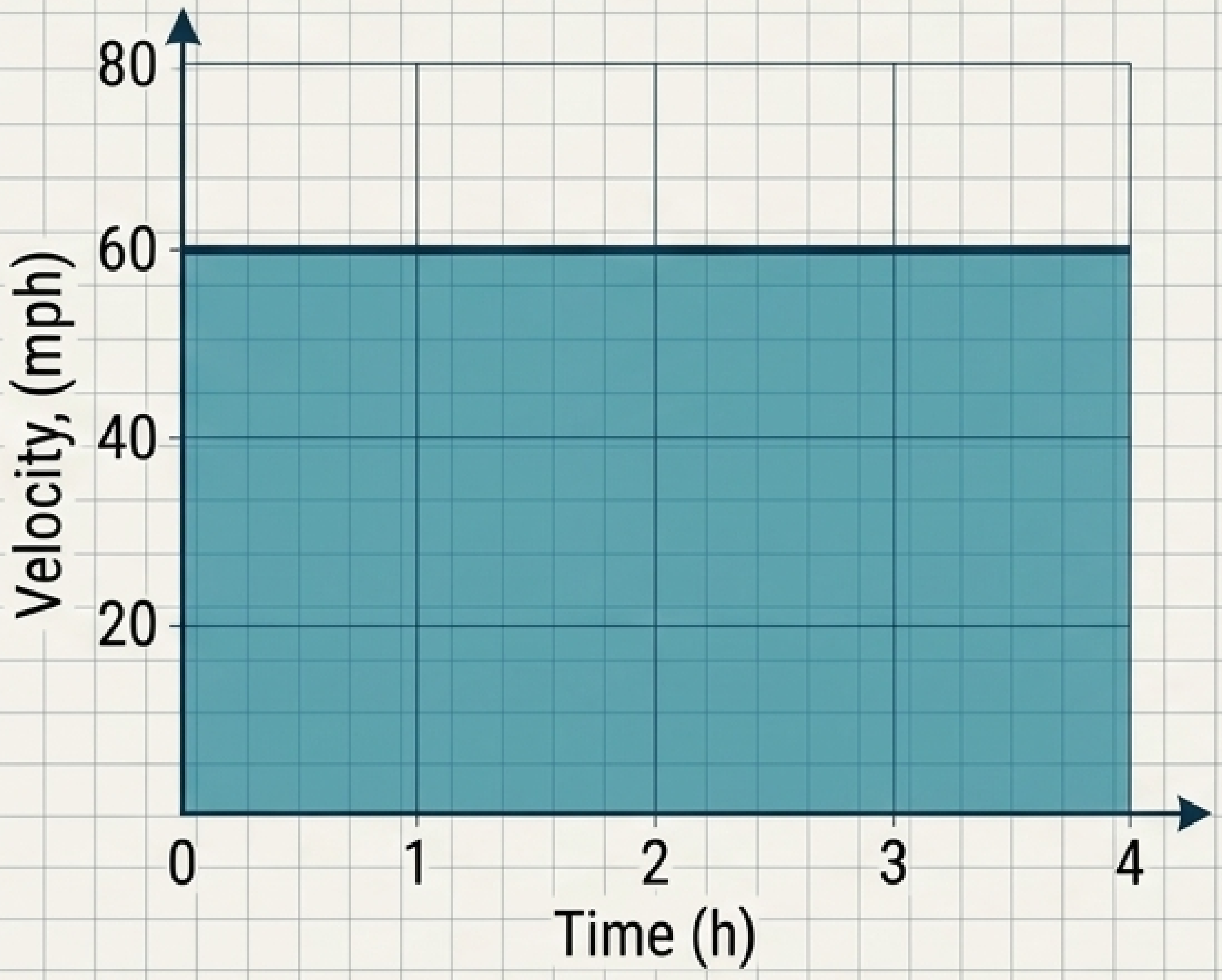


Sums & Sigma Notation

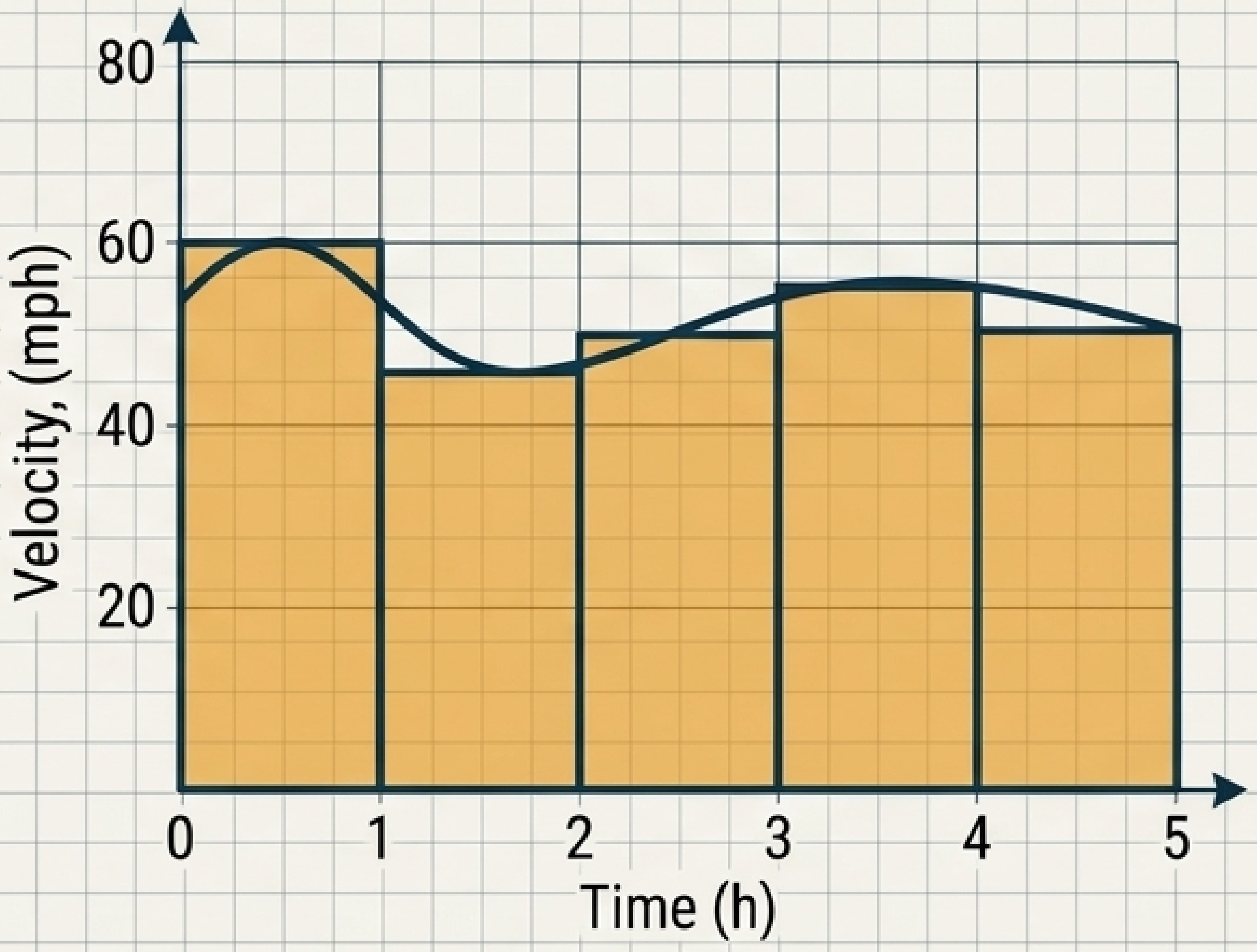
The Architectural Blueprints of Calculus

Constant Velocity



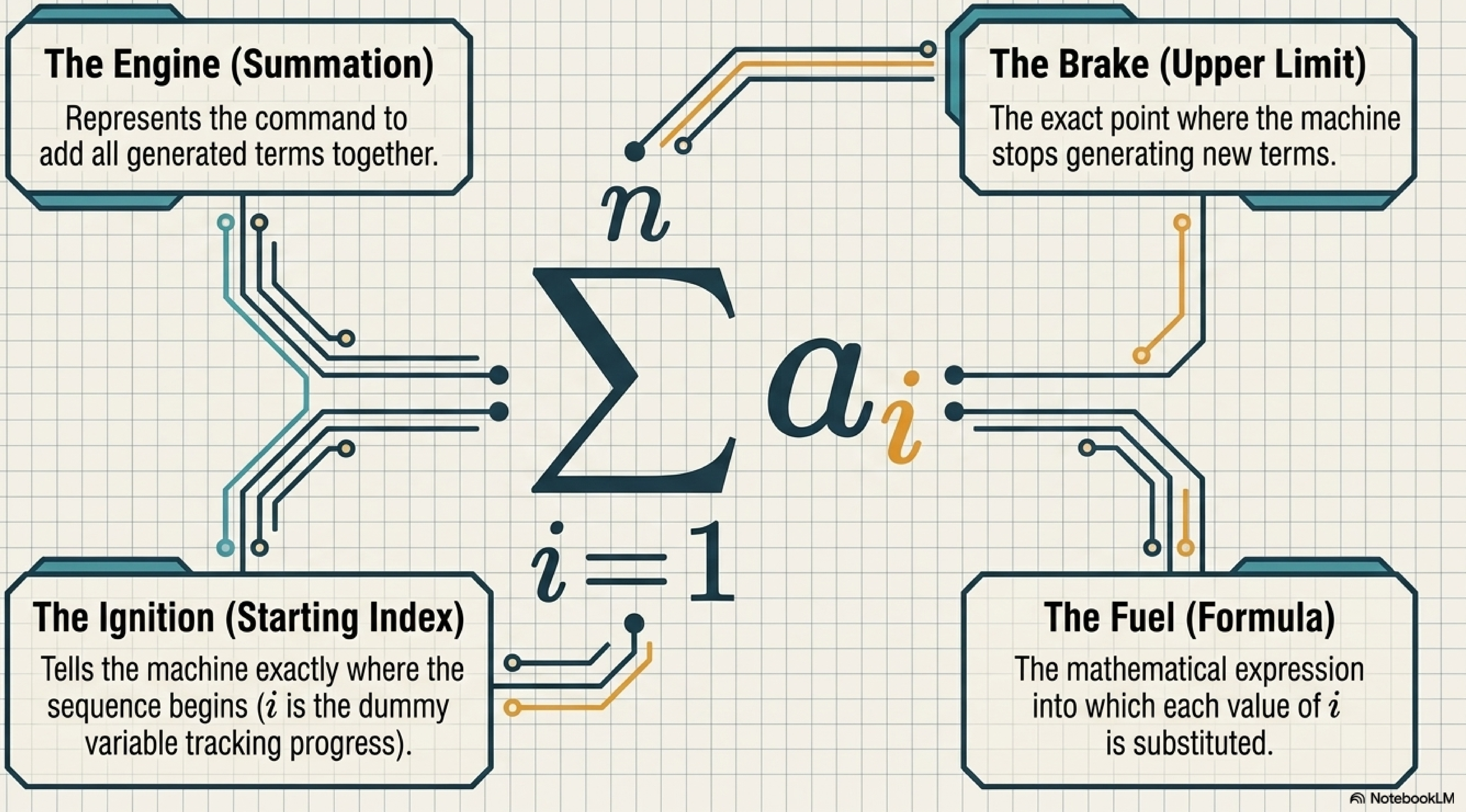
Distance is simply the area of one rectangle:
$d = r \times t$ (60 × 4 = 240 miles).

Varying Velocity



Area must be approximated by adding individual rectangles: $A \approx 60 + 45 + 50 + 55 + 50 = 260$ miles.

Adding 5 rectangles is trivial. But to get an accurate area, we need 5,000 infinitesimal rectangles. Brute-force addition is no longer possible. We need a machine to automate summation.



The Engine (Summation)
Represents the command to add all generated terms together.
The Brake (Upper Limit)
The exact point where the machine stops generating new terms.
$\sum_{i=1}^{n} a_i$
The Ignition (Starting Index)
Tells the machine exactly where the sequence begins (i is the dummy variable tracking progress).
The Fuel (Formula)
The mathematical expression into which each value of i is substituted.
NotebookLM

Human Instruction	Expanded Sequence	Sigma Blueprint
Sum the squares of the first 20 positive integers.	$1^2 + 2^2 + 3^2 + \ldots + 20^2$	$\sum_{i=1}^{20} i^2$
Sum the first 200 positive odd integers.	$1 + 3 + 5 + \ldots + 399$	$\sum_{i=1}^{200} (2i - 1)$
Sum the square roots of integers from 1 to 10.	$\sqrt{1} + \sqrt{2} + \sqrt{3} + \ldots + \sqrt{10}$	$\sum_{i=1}^{10} \sqrt{i}$

Key Insight: The dummy variable i acts purely as a placeholder to generate the sequence, drastically reducing the notation required.

The Historical Epiphany: At age 10, Carl Friedrich Gauss was tasked with adding all numbers from 1 to 100. Rather than brute-force counting, he invented a visual shortcut.

$(n+1)$ $(n+1)$ $(n+1)$

$$\text{A: } 1 + 2 + 3 + \dots + (n-1) + n$$

$$\text{B: } n + (n-1) + (n-2) + \dots + 2 + 1$$

$(n+1)$ $(n+1)$ $(n+1)$ $(n+1)$

By stacking and adding vertically, Gauss created n pairs that all equal $(n+1)$. Since he used two sequences, he simply divided by 2.

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

Theorem 2.1: The Summation Toolkit

Rule Name	Mathematical Blueprint	Visual Micro-Example
Sum of a Constant	$\boxed{\sum_{i=1}^{n} c = cn}$	$\sum_{i=1}^{5} 3 = 3 + 3 + 3 + 3 + 3 = 15\ (5 \times 3)$
Sum of First n Integers	$\boxed{\sum_{i=1}^{n} i = \frac{n(n+1)}{2}}$	$\sum_{i=1}^{4} i = 1 + 2 + 3 + 4 = 10 \rightarrow \frac{4(5)}{2} = 10$
Sum of Squares	$\boxed{\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}}$	$\sum_{i=1}^{3} i^2 = 1^2 + 2^2 + 3^2 = 14 \rightarrow \frac{3(4)(7)}{6} = 14$

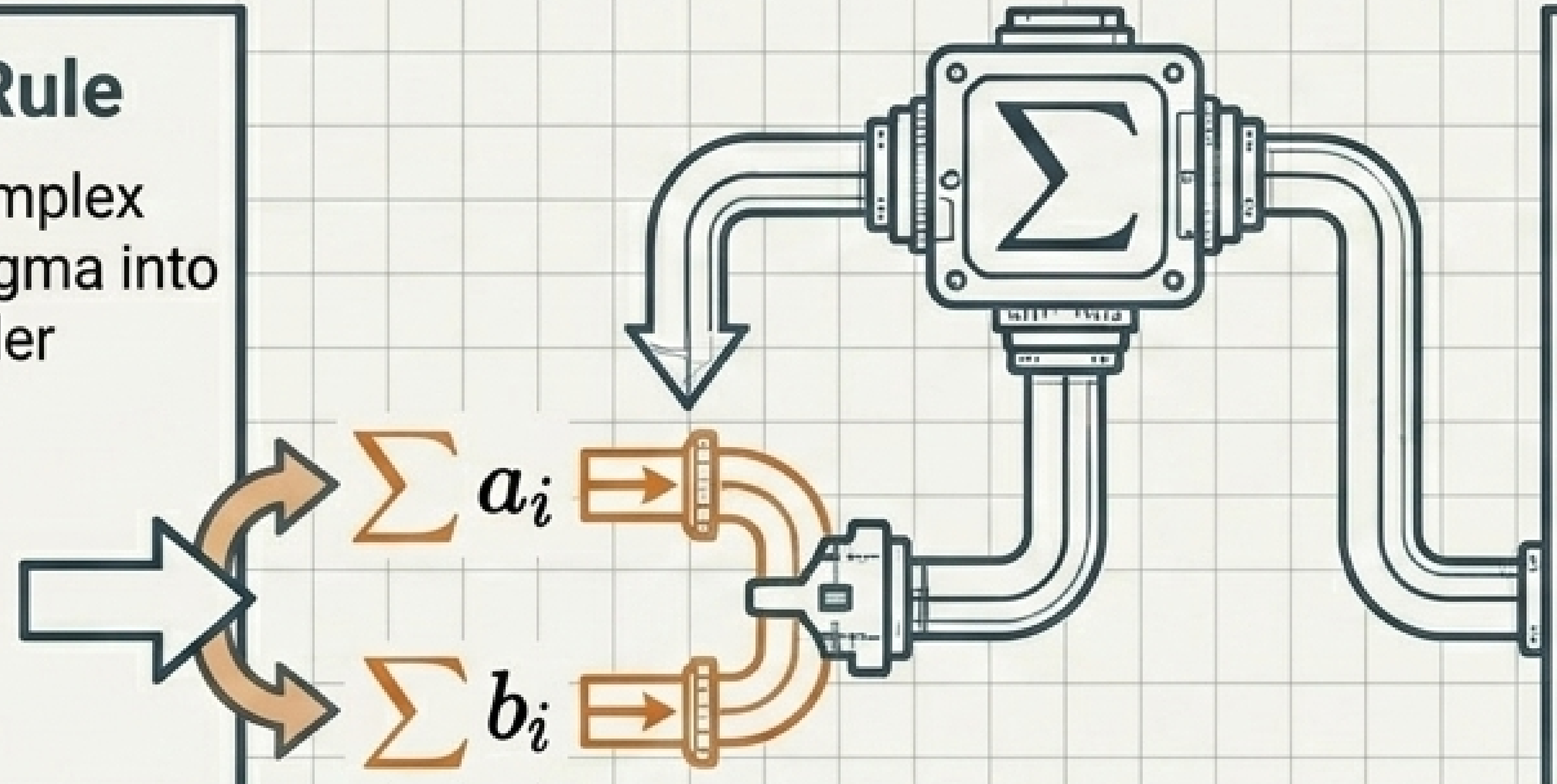
Theorem 2.2: The Rules of Linearity

$$\sum_{i=1}^{n}(ca_i + db_i) = c\sum_{i=1}^{n}a_i + d\sum_{i=1}^{n}b_i$$

The Splitting Rule

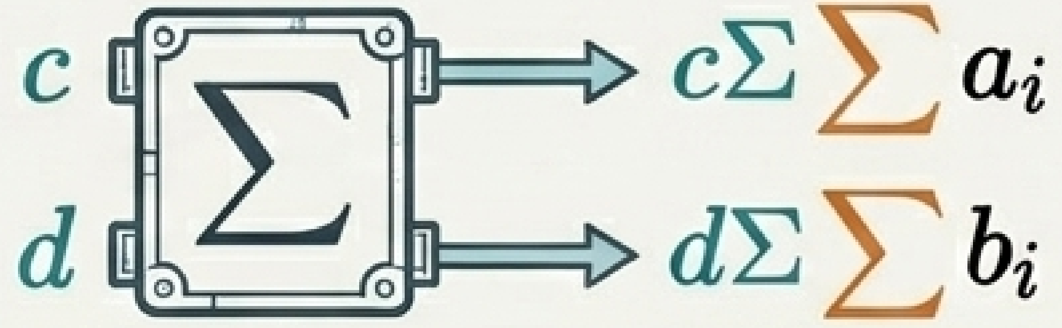
You can sever a complex addition inside a Sigma into two separate, smaller sums.

$$(a_i + b_i)$$



The Extraction Rule

Constants (c and d) act as multipliers. They can be safely pulled entirely outside the Sigma engine without altering the mathematics.



Core Principle: We never need to calculate complex expressions directly.
We dismantle them until they match our basic Toolkit.

Putting the Rules to Work

Target: Evaluate $\sum_{i=1}^{8}(2i+1)$

1. Split (Linearity)

Break the sum at the plus sign.

$$\rightarrow \sum_{i=1}^{8}(2i) + \sum_{i=1}^{8}(1)$$

2. Extract (Linearity)

Pull the constant multiplier '2' to the outside.

$$\rightarrow 2\sum_{i=1}^{8}(i) + \sum_{i=1}^{8}(1)$$

3. Apply Toolkit (Theorem 2.1)

Replace the Sigma notations with their Gauss formulas.

$$\rightarrow 2\left[\frac{8(9)}{2}\right] + (1)(8)$$

4. Solve

Standard arithmetic yields the final result.

$$\rightarrow 72 + 8 = 80$$

Bridging to Calculus: Evaluating Functions

Find the sum of values for $f(x) = x^2 + 3$ at intervals from $x = 0.1$ to $x = 1.0$.

Phase 1: Define the Interval

Values: $0.1, 0.2, 0.3 \ldots 1.0.$

We have 10 steps.
Therefore, the index runs from $i = 1$ to $n = 10$.
The interval multiplier is 0.1.

Phase 2: Construct the Blueprint

Substitute $x = 0.1i$ into the function:
$a_i = f(0.1i) = (0.1i)^2 + 3.$

Write the complete Sigma blueprint:

$$\sum_{i=1}^{10} \left[(0.1i)^2 + 3\right]$$

Phase 3: Execute the Machinery

Extract the constant:

$$\rightarrow 0.01 \sum_{i=1}^{10} i^2 + \sum_{i=1}^{10} 3$$

Apply Formulas:

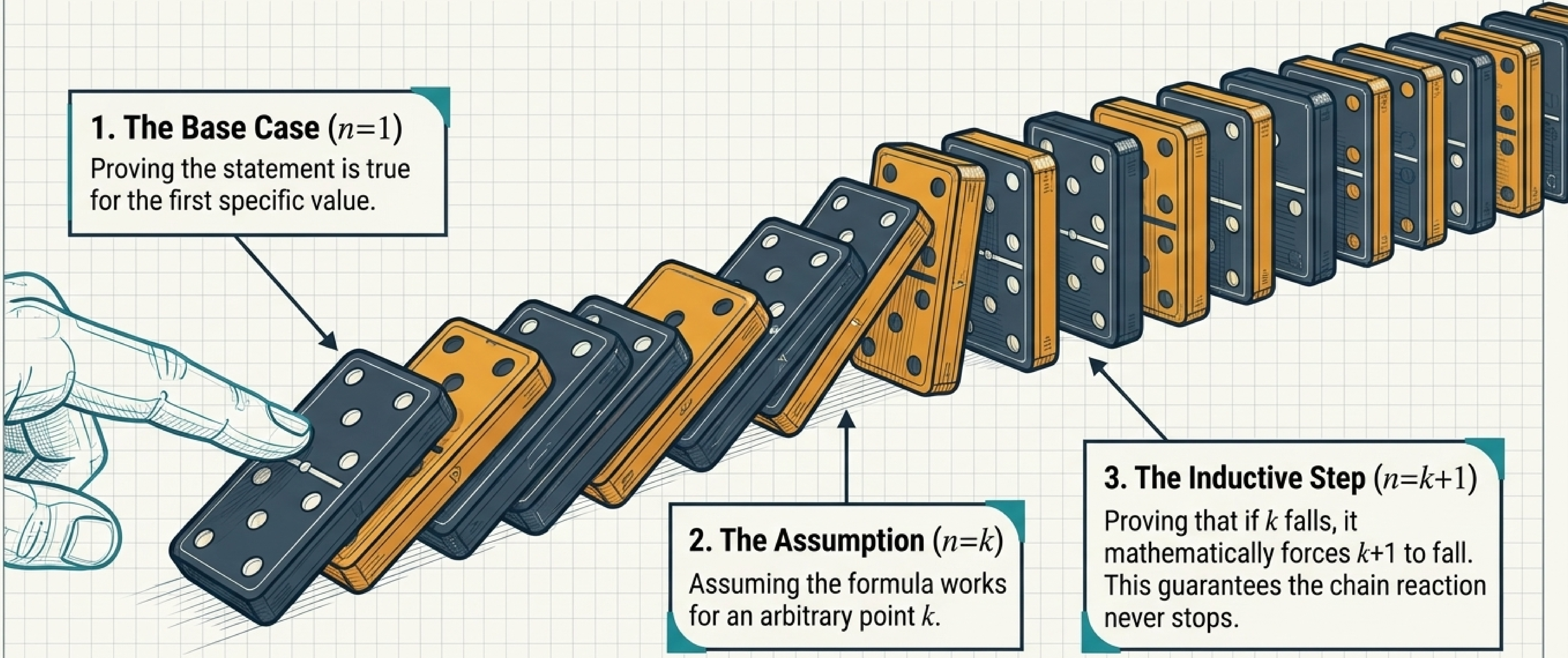
$$\rightarrow 0.01 \left[\frac{10(11)(21)}{6}\right] + 3(10)$$

$$= 33.85$$

Key Takeaway: We have successfully automated the calculation of 10 distinct geometric data points simultaneously.

The Engine of Proof: Mathematical Induction

How do we prove a formula works for infinite cases without checking them all manually? We use the domino effect.



Executing the Proof: Sum of Squares

Prove $\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$

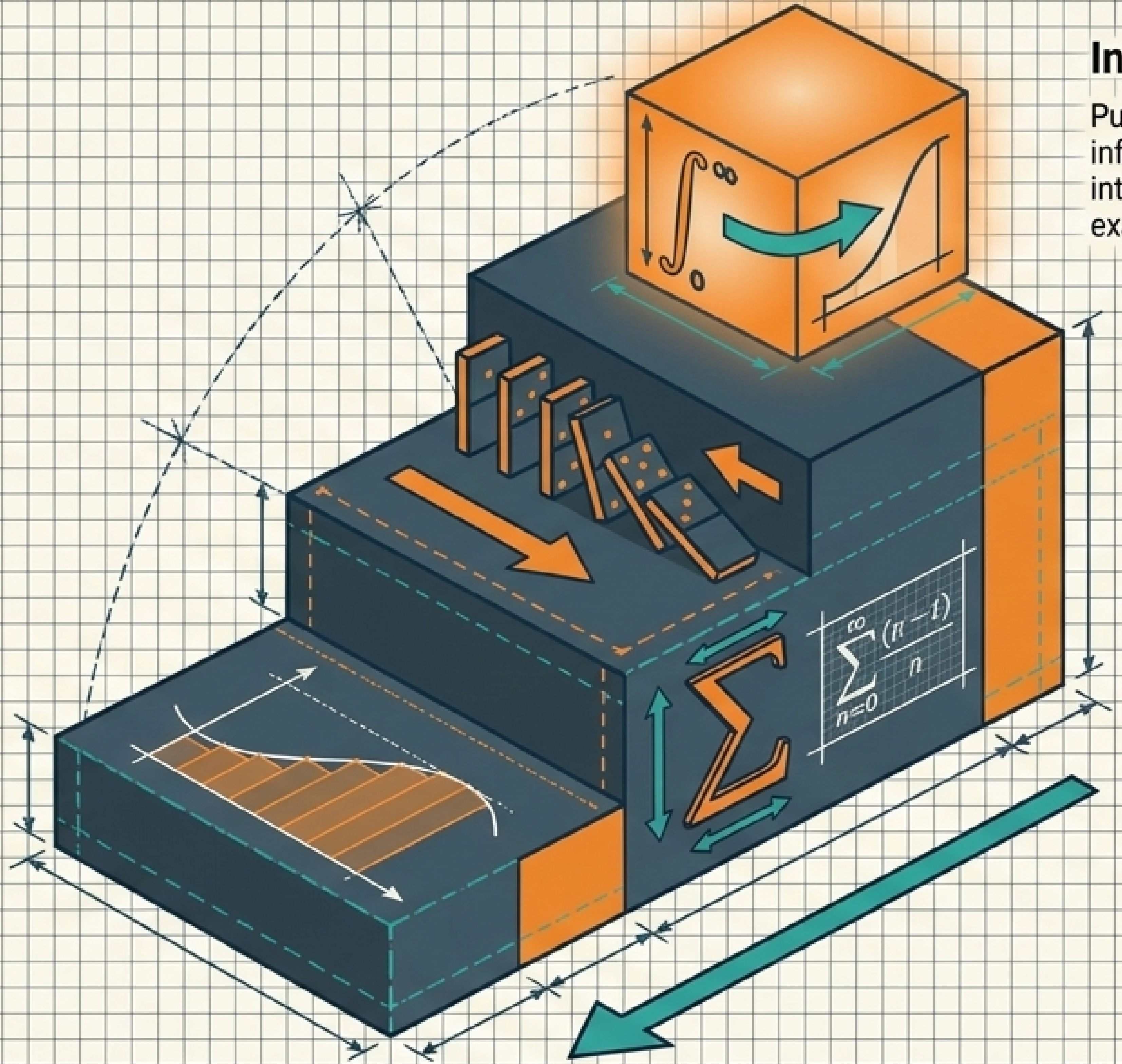
Algebraic Proof	Domino Phase Notes
Base Case ($n = 1$): $1^2 = \frac{1(2)(3)}{6} = 1$ ✓	**Phase 1:** The first domino falls.
Assume for $n = k$: $\boxed{\sum_{i=1}^{k} i^2 = \frac{k(k+1)(2k+1)}{6}}$ Inductive Hypothesis	**Phase 2:** Assume a domino falls at point k.
Inductive Step ($n = k+1$): $\sum_{i=1}^{k+1} i^2 = \boxed{\sum_{i=1}^{k} i^2} + (k+1)^2$ $= \frac{k(k+1)(2k+1)}{6} + (k+1)^2$	**Phase 3:** By finding a common denominator and factoring out $(k+1)$, this simplifies perfectly to the target formula for $k+1$. The chain reaction is mathematically proven!

Integral Calculus

Pushing $n \to \infty$. As rectangles become infinitely thin, discrete sums transform into continuous integrals, yielding exact area.

The Logical Foundation

Securing the machinery via Mathematical Induction, ensuring formulas hold true to infinity.



The Notation & Shortcuts

Deploying Sigma (Σ) and Gauss's Toolkit to automate massive, repetitive additions.

The Physical Problem

Estimating the area under curves using discrete rectangular slices.

Closing Insight: Sigma notation isn't just a shorthand for arithmetic; it is the vital architectural bridge between discrete geometry and modern calculus.